

Mahesh Bhandari

5th Sem Computer Engineering · Pulchowk Campus, TU | Full-Stack & ML/AI
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TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python, SQL, Bash

Backend & Databases: Node.js, Express.js, MongoDB, REST API Design, FAST API, Django

AI / ML: PyTorch, Transformer architectures, RAG pipelines, vector embeddings

DevOps & Tools: Docker, Kubernetes (k8s), GitHub Actions, Git, Postman

PROJECTS

GPT-3 Transformer — From Scratch | Python · PyTorch · LLM 2025

- Built a 355M-parameter Transformer from scratch following the "Attention is All You Need" paper, implementing multi-head self-attention, positional encoding, and layer normalization without any pre-built LLM libraries.
- Reduced training loss by ~60% over 10k iterations through optimized vectorized PyTorch training loops and careful learning rate scheduling.
- Demonstrates deep understanding of LLM internals — the kind of project most developers only read about.

YouTube Video Chat — RAG Q&A System | Python · LangChain · Vector DB 2025

- Built an end-to-end RAG pipeline that ingests YouTube video transcripts, chunks and embeds them into a vector store, and enables natural language Q&A over video content.
- Handles multi-turn conversational queries with context retrieval, reducing irrelevant answers compared to a naive keyword search approach.
- Usable on any public YouTube video — zero manual labeling or data preprocessing required.

Sikshya Kendra — EdTech Platform | Freelance · Full-Stack 2025

- Delivered a full educational platform for a paying client, supporting online courses, live classes, and in-app tutoring across multiple subjects.
- Designed the resource hub and course discovery flow, reducing time-to-enrollment for students.
- Deployed and maintained the platform end-to-end, handling client feedback rounds and iterative updates.

Room-Bazar — Room Rental Marketplace | MERN Stack 2024

- Built a full-stack rental platform for Kathmandu and Pokhara, digitizing a process that was previously handled offline via word-of-mouth.
- Implemented JWT-based cookie authentication, real-time booking, and instant messaging between tenants and landlords.
- Advanced filtering (price, location, amenities) and live updates significantly improved the room-search experience over static listing sites.

Tweet-Tube — Unified Social API | MongoDB · Express.js 2024

- Engineered a RESTful backend combining Twitter-style short posts with YouTube-style video metadata into a single unified API — 30+ documented endpoints.
- Comprehensive Postman collection covers happy-path and edge-case scenarios; API documentation published for open use.

ACHIEVEMENTS & LEADERSHIP

Instructor — Software Fellowship Program | Locus 2026 Pre-Event, IOE Pulchowk Jan 2026

- Selected to teach foundational software engineering skills to a cohort of junior students as part of the pre-event for Nepal's largest tech festival — one of very few 5th-semester students chosen as an instructor.

Hack-A-Week Finalist | Locus 2026, IOE Pulchowk March 2026

- Reached the finals of Hack-A-Week, a week-long competitive hackathon held at Locus 2026, competing against teams from across Tribhuvan University.

EDUCATION

BE in Computer Engineering — Pulchowk Engineering Campus, Tribhuvan University *2023 – Present*
Currently in 6th semester. Pulchowk is Nepal's top-ranked engineering campus.

+2 in Computer Science — Himalayan Whitehouse International College *2021 – 2023*
GPA: 3.86 / 4.0